Notes on “The Dawn of Robot Surveillance” by Jay Stanley

The Rise of AI-Powered Cameras

Past: Dumb Cameras

Traditional surveillance cameras simply recorded video, requiring human review to extract insights.

Present: Basic Analytics

Thanks to the “revolution” in “deep learning” and the shift to storing information in the cloud, current AI-powered surveillance systems can detect all kinds of things pertaining to people, places, and objects.

Future: Intelligent Monitoring

AI-powered cameras will be able to understand far more complex human behaviors and make advanced, autonomous decisions.

Phase Transition in Surveillances

“We are on the cusp of a fundamental change in the nature of surveillance — akin to a “phase transition” in physics. Lower the temperature of a glass of water from 40 to 35 degrees and you get slightly cooler water — a minor, linear change. But cool your water from 35 to 30 degrees and you get a radical qualitative change in the nature of your water. That kind of discontinuous phase transition is what we are facing as we move from collection-and-storage surveillance to mass automated real-time monitoring.”

Stanley Asks Us to Imagine

A possible future

"Here's a list of enemies of my administration. Have the cameras send us all instances of these people kissing another person on the lips, and the IDs of who's in them."

Stanley Asks Us to Imagine

A possible future

"Have the cameras send us more on that suspect, including everybody he's been seen in public with for the last 18 months, and the AI's assessment of his relationship with each."

Stanley Asks Us to Imagine

A possible future

"Our cameras can now provide police chiefs a daily list of all the people in an area who were intoxicated in public the night before, based on changes in their normal gait, speech patterns, and remote physiological measurements."

Stanley Asks Us to Imagine

A possible future

"Hi there, we saw you jogging by at a 7:28/mile pace earlier today. We have a special life insurance offer just for healthy people like you!"

Stanley Asks Us to Imagine

Video analytics: a scenario

Robert is on his way home from work and stops at a shopping mall to run an errand. His day ended badly after a coworker was rude. As he walks through the mall, a camera notices that he looks angry. His eye movements, heart rate, and "body affect" back that up. The camera notes that he's a young male. He's wearing clothing that is fashionable at his school, but which the camera has decided on its own is correlated with anti-social behavior. The mall's auto-PTZ cameras swivel to follow and zoom in on him. The system parses the text on his t-shirt and the symbols of a tattoo on his arm and decides those are also "risk indicators." An alert flags him for security staff. As he prepares to walk back outside he pulls his gloves out of his pocket and, unnoticed by him, a paper receipt falls out of his pocket. The zoomed PTZ camera catches it, and sounds an "abandoned object" alarm.

Robert finds himself being confronted by mall security staff and accused of littering. He's annoyed at being hassled and can't hide it. The security staff, partly influenced by his automated security rating, get belligerent with him and tell him he's never to set foot in the mall again, even though he's been going there ever since he was a small child. His photo, body dimensions, and gait pattern are entered into the camera system and an alarm will sound if it ever spots him again — at that mall or on the premises of the owner's other corporate properties and partners.

He appeals, saying it was an accident, but can't get the mall to share the video footage it has of the "littering." Or, of how the staff treated him. His friends and schoolmates hear about what happened and become very self-conscious when they enter the mall, and the fast-growing list of other places where they've learned that similar technology is being deployed. Some of them find themselves dressing more conservatively to blend in better and lessen the chances of being hassled.

As this kind of scenario is repeated tens of thousands of times across the nation, the atmosphere of our public spaces and the nature of American life slowly becomes less vibrant.

Key Capabilities of AI Video Analytics

Human Action Recognition

Detecting specific behaviors like running, fighting, or loitering.

Anomaly Detection

Flagging unusual or suspicious activities.

Emotion Recognition

Analyzing facial expressions and body language to infer emotional states.

Demographic Analysis

Estimating age, gender, ethnicity of individuals.

Current datasets used for human activity recognition training include thousands of terms.[52] Examples include:

Drink water	Clapping	Tear up paper	Take off jacket
Put on hat	Take off glasses	Hand waving	Reach into pocket
Phone call	Playing with phone	Taking selfie	Wipe face
Staggering	Vomiting	Punch/slap	Pat on the back
Touch pocket	Walking towards	Handshaking	Walking apart
Play guitar	Smoking a cigarette	Use laptop	Walking the dog
Fan self	Drinking coffee	Hugging	Drinking beer

Current Deployments

Government

- Law enforcement agencies are using video analytics for crime prevention and investigation. For example, New York’s Domain Awareness System and school security systems.

Commercial

- Retailers are adopting AI video to detect shoplifting and analyze customer behavior. Transportation hubs use it to monitor crowds and detect suspicious activities.

The Promise of Video Analytics

Enhanced Public Safety

- Faster response to emergencies and crime prevention.

Less Labor

- Automated monitoring of large areas with less manpower.

Business Intelligence

- Greater insights into customer behavior and preferences.

Chilling Effects

Constant AI monitoring could lead to self-censorship and behavioral changes as people become aware that their every move is being scrutinized and evaluated. This could have a significant impact on freedom of expression and association.

Awareness of Surveillance

Self-Consciousness

Behavioral Changes

Reduced Freedom

New Types of Data Collection

Clothing and Appearance

- Analyzing fashion choices and tattoos to determine if someone is dressed in an “anomalous” way—clothes are too expensive, too cheap, too religious?

Biometric Data

- Gait analysis, heart rate, and other physiological measurements

Social Interactions

- Detecting relationships and group dynamics.

Unreliable Analytics

Many video analytics capabilities are still unproven or based on dubious science. Emotion recognition, for example, often relies on oversimplified assumptions about the relationship between facial expressions and internal states.

False Positives

- Innocent people may be flagged as suspicious due to inaccurate algorithms.

Pseudoscience

- Some analytics attempt to infer internal states or intentions, which is not scientifically valid.

Discriminatory Effects

AI video analytics systems embedded in everything from street cameras to body cameras may incorporate biases that lead to discriminatory outcomes, even if not explicitly programmed to do so. This could exacerbate existing inequalities in law enforcement and public spaces.

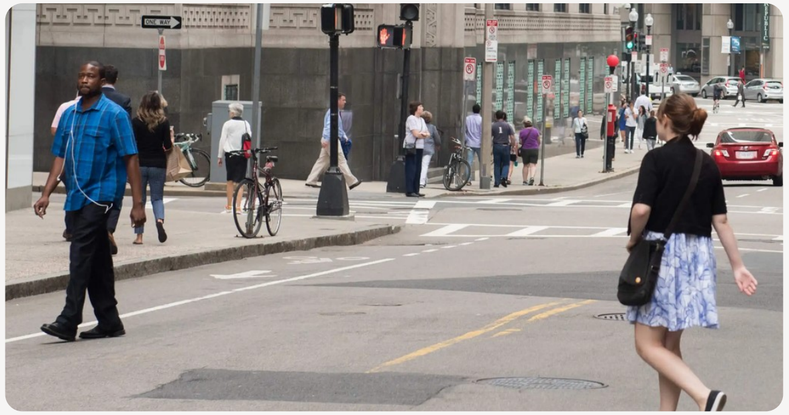
Algorithmic Bias

Disproportionate Targeting

Amplified Discrimination

Over-Enforcement

AI-powered cameras could enable the strict enforcement of minor infractions that are currently often overlooked. This could lead to a more rigid and less forgiving society.



Petty Violations

- Jaywalking, littering, or minor traffic infractions could be constantly monitored and penalized.

Context

- AI systems may not account for context or extenuating circumstances.

Increased Fines

- More violations detected could lead to more fines and penalties.

Potential for Abuse

Powerful surveillance systems create opportunities for misuse by those who control them. Historical examples show how surveillance has been used to target political dissidents, minorities, and other vulnerable groups.

Political Repression

- Tracking and suppressing activists or opposition groups.

Blackmail

- Using collected data to coerce or manipulate individuals.

Corporate Espionage

- Spying on competitors or whistleblowers.

Constitutional Concerns

Privacy

- The collection of physiological data or comprehensive tracking of movements could infringe on reasonable expectations of privacy.

Fourth Amendment Violations

- Warrantless collection of personal data through AI analytics may constitute unreasonable searches.

First Amendment Impact

- Pervasive surveillance creates chilling effects on free speech and freedom of association.

Recommendations: Democracy and Transparency

Public Approval

- Require legislative approval for government deployment of video analytics.

Impact Assessments

- Conduct and publish civil rights and civil liberties impact assessments.

Transparency

- Disclose types of data collected and allow individuals to access their data.

Oversight

- Implement robust auditing and independent testing mechanisms.

Recommendations: Protections from Misuse

Narrow Purpose

- Limit deployments to specific, clearly defined purposes.

Proportionality

- Ensure the value of use is proportional to privacy risks.

Data Minimization

- Collect and retain only necessary data.

Privacy Best Practices

- Follow best practice data management guidelines.

Recommendations : Decision Making

Human Oversight

- Prohibit autonomous decisions that have significant legal implications.

Fairness and Transparency

- Ensure equitable and explainable decision-making processes.

Clear Rules

- Only flag behavior that violates clearly established guidelines.

Private Sector Considerations

Ethical Use

- Companies should adhere to the same strictures as government when deploying video analytics that affect the public.

Transparency

- Businesses must be open about their use of AI surveillance and data collection.

The Global Context

While Stanley’s report focuses on the U.S., there are global implications of AI video analytics.

China

- Extensive use of AI surveillance.

Russia

- Expanding facial recognition and monitoring systems

Singapore

- Smart city initiatives incorporating AI-powered cameras